

Resistance-aware computational design of CRISPR gene-drive target sites in the *Anopheles gambiae* doublesex locus

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Abstract

Malaria kills over half a million people annually, and CRISPR homing gene drives that render *Anopheles* mosquitoes refractory to parasite transmission are among the most discussed genetic vector-control strategies. Their two dominant failure modes are weak guide-RNA efficacy at the drive target site and resistance alleles generated by end-joining repair. We present a fully computational design study of the *Anopheles gambiae* doublesex (*dsx*, AGAP004050) locus - the target of the only gene drive shown to collapse caged mosquito populations (Kyrou et al. 2018) - that couples a convolutional neural network (CNN) trained on 5,310 experimentally measured CRISPR guides (Doench 2016) with a resistance-aware composite ranking and a quantitative model of drive/resistance allele dynamics. Scanning a 110 kb region around the locus (NC_064601.1:47600000-47710000) recovered 10,936 NGG target sites, of which 9,328 unique sites lie in the *dsx* gene body. Our pipeline recovers the published, experimentally validated Kyrou *dsx* gRNA as a positive control (score 1.137, zero predicted off-targets within the locus) and ranks 93 candidate sites above it, led by ACGGACGGATACAGACTGGA (CNN score 1.345). A discrete-generation recursion over homing rate h and resistance-generation rate e shows that under full homozygous sterility, resistance alleles dominate within 60 generations for every parameter combination tested; the drive exceeds 95% allele frequency only when $h = 0.99$ and $e \leq 0.001$. Target-site choice is therefore primarily a resistance-engineering problem, not an efficacy problem: sites where any indel destroys gene function (such as the essential *dsx* exon-5 splice junction) are the only viable drive targets. A constraint-aware second ranking integrates exon and splice-junction annotations and independently concentrates on the functional region the field validated. We release ranked designs, the full simulation grid, six figures, a 22-test verification suite, and all code.

1. Introduction

1.1 Malaria and the limits of conventional vector control

Malaria remains one of the largest infectious-disease burdens on earth. The World Health Organization estimates roughly a quarter of a billion cases and over half a million deaths per year, with the great majority of mortality concentrated in sub-Saharan Africa and in children under five. The dominant transmission vectors are mosquitoes of the *Anopheles gambiae* complex, and the two pillars of vector control - long-lasting insecticidal nets (LLINs) and indoor residual spraying (IRS) - are responsible for most of the gains of the last two decades. Both pillars are now eroding.

Pyrethroid resistance is widespread in *An. gambiae* populations across Africa, behavioural shifts (outdoor and early-evening biting) reduce contact with indoor interventions, and the logistical and financial cost of sustaining near-universal net coverage is high. Progress against malaria has stalled since roughly 2015, and the search for complementary, self-sustaining interventions has become a central priority of the field.

Genetic vector control offers that complementary axis. Rather than killing mosquitoes with an external agent that selects for resistance and must be reapplied indefinitely, genetic strategies modify the mosquito population itself - either suppressing it (population suppression) or rendering it unable to transmit the parasite (population replacement). Earlier genetic approaches, such as the sterile insect technique and releases of males carrying a dominant lethal, require sustained mass releases because the modified insects disappear from the population. A gene drive, by biasing its own inheritance, can in principle spread from a single small release and persist, which changes the economics of intervention fundamentally.

1.2 CRISPR homing gene drives

A CRISPR homing gene drive encodes the Cas9 nuclease and a guide RNA (gRNA) inside the genomic site that the gRNA targets. In a heterozygote carrying one drive allele and one wild-type allele, Cas9 cuts the wild-type chromosome at the target site; when the break is repaired by homologous recombination using the drive chromosome as template, the drive allele is copied onto the wild-type chromosome. An organism that would transmit the drive to 50% of its offspring instead transmits it to a fraction approaching the homing rate, so the allele spreads super-Mendelianly even when it carries a fitness cost (Burt 2003). The first laboratory demonstrations came in 2015: Gantz and Bier showed a working CRISPR drive in *Drosophila*, and Gantz et al. built a drive in *Anopheles stephensi* carrying anti-parasite effector genes. Hammond et al. (2016) extended the approach to *An. gambiae* with drives targeting three genes required for female fertility.

The Hammond study also exposed the central failure mode. Its drives cut their targets efficiently, but within a few generations the drive frequency stalled and resistant alleles - small insertions and deletions produced when the cut was repaired by end-joining rather than homologous recombination - accumulated and ultimately dominated. Many of these indels restored a functional, drive-resistant target sequence. Unckless and colleagues formalized the problem theoretically in 2017: a CRISPR drive in a panmictic population is mathematically unstable to resistance unless the rate at which functional resistant alleles are generated is very low, and the equilibrium outcome depends far more on the resistance-generation rate than on the homing rate. The design implication is that the gRNA target sequence must be chosen so that essentially any mutation at the site destroys the function of the gene - in which case resistant alleles are nonfunctional and are purged by selection rather than spreading.

1.3 The doublesex target

Kyrou et al. (2018) identified a target with exactly that property. The *doublesex* (*dsx*) gene sits at the bottom of the insect sex-determination cascade: it is a transcription factor of the DM-domain family, conserved across insects, whose sex-specific splice isoforms direct male versus female somatic development. In *An. gambiae*, the female isoform (*dsxF*) incorporates a female-specific exon (exon 5), while the male isoform (*dsxM*) splices past it; the decision is controlled upstream by the splicing factor transformer-2, but *dsx* is the final effector. Females homozygous for a disrupted *dsxF* develop as intersex - sterile, with male-like morphology, and unable to blood-feed - while males carrying the same disruption are essentially unaffected, because *dsxM* does not use the targeted exon. This asymmetry is what makes the locus a suppression target: a drive inserted at the female-specific exon boundary imposes its fitness cost only on females, and only when homozygous, so heterozygous carriers of both sexes transmit the drive normally while homozygous females are removed from the reproductive pool. Crucially, the intron-4/exon-5 boundary region is highly conserved and functionally constrained: indels that destroy the drive target also destroy *dsx* function, so resistance alleles generated at the site are sterile and removed by selection. A drive inserted at this boundary spread to fixation in caged *An. gambiae* populations and drove them to collapse within 8-12 generations, with no functional resistance emerging over the course of the experiments. It remains the only CRISPR drive to have collapsed caged mosquito populations, and its success is the strongest empirical evidence that target-site choice - not nuclease engineering - is the decisive variable in drive design.

1.4 The design problem

A practical drive target design must jointly optimize three quantities: (i) the on-target cleavage efficacy of the gRNA, which sets the homing rate; (ii) the off-target load, which sets the rate of collateral mutation; and (iii) the rate at which functional resistance alleles are generated at the target, which is a property of the sequence context and the functional constraint acting on it. Public gRNA design tools address (i) and partially (ii). Sequence-based tools such as CHOPCHOP and CRISPOR rank guides by predicted efficacy and off-target counts; the Azimuth/CRISPick model of Doench et al. (2016) provides a strong sequence-only efficacy predictor trained on thousands of experimentally measured guides; CRISPROff predicts energy-based off-target binding. None of these tools models resistance generation, none integrates a population-genetic simulation of the resulting drive, and none ranks candidate sites within a specific target locus against an experimentally validated positive control. The consequence is that a designer who wants "a *dsx*-like target" has no quantitative pipeline: the successful choice in Kyrou et al. was reached through biological insight and manual curation, not through an automated, reproducible scoring procedure.

1.5 Hypotheses

This study builds that missing pipeline for the *dsx* locus and quantifies the drive-dynamics consequences of each design axis. We test three hypotheses.

H1 (detection): an NGG scanner over the true *dsx* locus recovers the published Kyrou gRNA with competitive predicted efficacy and low predicted off-target load.

H2 (ranking): machine-learned efficacy scoring identifies *dsx* gene-body sites predicted to outperform the published gRNA.

H3 (dynamics): across a parameter grid, the resistance-generation rate e - not the homing rate h - is the binding constraint on long-term drive success under a sterility-inducing drive.

2. Methods

2.1 Data and locus identity

The 110 kb locus region NC_064601.1:47600000-47710000 (idAnoGambNW_F1_1 assembly) was fetched live from NCBI E-utilities (efetch, 24 Sep 2026) and archived as data/dsx_locus_region.fasta. Locus identity was verified through NCBI Gene 1270904, whose record lists aliases AGAP004050 and DSX and designations including "female-specific doublesex protein" (evidence: data/dsx_gene_esummary.json). The RefSeq annotation of the locus, fetched as the NCBI gene table (data/dsx_gene_table.txt), contains nine transcript variants whose exon models were parsed for constraint-aware scoring. The published Kyrou gRNA target plus PAM (5'-GTTTAACACAGGTCAAGCGG TGG-3') was taken from the article full text (PMC6871539). All network access is confined to scripts/fetch_data.sh; every analysis and every test runs against the archived local copies.

2.2 Target-site scanning

Every 23-nt window on both strands of the 110,001-nt region was tested for an NGG PAM; matching windows yield a 20-nt protospacer, its strand, its position, and a 30-nt CNN context sequence (4 bp upstream + protospacer + PAM + 3 bp downstream, N-padded at region edges). Windows containing ambiguous bases are rejected. Gene-body sites are defined by overlap with the *dsx* transcription span (minus strand, 47,610,877-47,700,920) and deduplicated so that each unique protospacer is scored once.

2.3 CNN efficacy model

Efficacy scoring uses a 1-D convolutional neural network over one-hot-encoded sequence context. Each 30-nt context window is encoded as a 4 x 30 binary matrix; the network also receives the GC fraction and the 16 dinucleotide frequencies of the protospacer as explicit engineered features. The architecture is:

stage	operation	output shape
input	one-hot 30-nt context	4 x 30
conv block 1	Conv1d(4-32, kernel 5) + BatchNorm + ReLU	32 x 30

conv block 2	Conv1d(32-64, kernel 5) + BatchNorm + ReLU	64 x 30
conv block 3	Conv1d(64-64, kernel 5) + BatchNorm + ReLU	64 x 30
pooling	global average over positions	64
head	concat pooled features + GC + 16 dinucleotide features; MLP to scalar	1

Training used the Doench 2016 FC+RES dataset (5,310 experimentally measured 30-mer guides with drug-gene rank scores; mirror of the Microsoft Research azimuth training data), split 80/20 into 4,248 training and 1,062 held-out guides with a fixed seed. The model minimized mean squared error with the Adam optimizer for 15 epochs (seed 0); no early stopping or hyperparameter search was applied, so the reported held-out performance is not selected on the test split. A ridge regression over the same sequence features serves as the classical baseline. Held-out performance: Spearman 0.556 for the CNN versus 0.463 for ridge. As an external check, both models were evaluated without retraining on the independent Doench V1 dataset (2,144 guides measured in a different experimental setup): 0.609 versus 0.444. The CNN's cross-dataset advantage over ridge (+0.166) is larger than its within-dataset advantage (+0.093), indicating that the convolutional features generalize across assay conditions rather than memorizing the training assay. We note the cross-species caveat: the CNN was trained on human-cell data, and its absolute scores transfer to mosquito only as a ranking prior (see Limitations). The checkpoint and metrics are archived in assets/ with full provenance.

2.4 Off-target counting

Candidate sites are screened against the full 110 kb region for near matches with up to three mismatches. Exact enumeration is avoided via a pigeonhole index. Each 20-mer is split into four exact 5-mer blocks. Any 20-mer within three mismatches of the query differs in at most three of its positions, so by the pigeonhole principle at least one of its four 5-mer blocks is exactly identical to the corresponding query block; a 5-mer inverted index over the region therefore enumerates a small candidate set containing every true ≤ 3 -mismatch match, and each candidate is verified exactly by direct comparison. The method returns identical counts to an exhaustive sliding-window scan on randomized test panels (test suite), so the speedup changes cost, not results.

2.5 Composite ranking (v1)

Sites are ranked by score = CNN_efficiency - 0.5 (*off-targets within 3 mismatches*) - 0.2 [GC fraction outside 0.3-0.8] - 0.1 * max(0, longest homopolymer run - 3). Penalty weights were fixed a priori, before any candidate list was inspected, so that the ranking cannot be tuned toward a preferred outcome. The top-150 candidates by raw CNN score are screened for off-targets and reranked by the composite.

2.6 Constraint-aware ranking (v2)

The v2 ranking adds a functional-constraint term derived from the union of the nine RefSeq transcript exon models: +0.25 if the majority of the 23-nt site is exonic, and a further +0.5 if any part of the site lies within 15 bp of an annotated exon boundary (splice-junction proximity). The term encodes the design lesson of section 1.3 - constraint, not raw efficacy, is what suppresses resistance - using only annotation data, fixed before the v2 ranking was computed.

2.7 Drive dynamics

Population consequences are evaluated with a three-allele model (W wild-type, D drive, R resistant) with discrete, non-overlapping generations, following the Burt/Unckless functional form. Let w , d , r be the adult allele frequencies entering a generation, with $w + d + r = 1$ (a conservation invariant the test suite checks at every simulated generation). Random mating produces diploid zygote frequencies from these allele frequencies under Hardy-Weinberg proportions. In the germline of W/D heterozygotes, homing converts the W allele to D with probability h ; of the conversions that fail, a fraction e yields a functional resistant allele R (end-joining repair that restores a cleavage-resistant but functional target), and the remainder yields nonfunctional products that we absorb into the sterile class. Gamete frequencies after conversion are therefore $g_D = d + h f_{WD} / 2$, $g_R = r + e (1 - h) f_{WD} / 2$, $g_W = w - h f_{WD} / 2 - e (1 - h) f_{WD} / 2$, where f_{WD} is the zygote frequency of W/D heterozygotes. Genotype fitnesses are 1 for wild-type homozygotes, $1 - s_{het}$ for drive heterozygotes, and $1 - c_{hom}$ for drive homozygotes, with $c_{hom} = 1$ modeling the full female sterility of the Kyrou drive; resistant alleles are wild-type in fitness. The next generation's allele frequencies are the fitness-weighted gamete frequencies renormalized by mean fitness. The Wright-Fisher implementation samples $2N = 20,000$ alleles each generation from the deterministic gamete frequencies (multinomial drift, five random seeds) to check robustness to genetic drift. The parameter grid spans h in $\{0.80, 0.90, 0.95, 0.99\}$ x e in $\{0.001, 0.01, 0.1, 0.5\}$, 60 generations, drive introduced at 1% allele frequency, $s_{het} = 0$. All parameter values are modeling choices, stated, and swept rather than tuned.

2.8 Verification

A 22-test hermetic pytest suite covers scanner correctness (both strands, edge padding, ambiguity rejection, homopolymer detection), the pigeonhole/naive off-target agreement, model output shapes and finiteness, composite-score penalty behavior, annotation parsing and the documented Kyrou annotation discrepancy, constraint-bonus logic, recursion conservation (allele frequencies sum to 1 at every generation), qualitative drive behavior (invasion from rare, no spread without homing, resistance blocking, Kyrou-like suppression), and seeded Wright-Fisher reproducibility. Live network calls exist only in `scripts/fetch_data.sh`, never in tests.

3. Results

3.1 Positive control

The scanner recovered the published Kyrou *dsx* gRNA exactly once in the locus (minus strand, region offset 22,174, genomic ~47,622,174), consistent with *dsx* being a minus-strand gene. Its CNN efficacy (1.137) lies in the upper tail of the 10,936-site distribution (mean 0.763, sd 0.178, range -0.042 to 1.345; Figure 1), and it has zero predicted off-targets within the 110 kb panel. H1 is supported. One annotation discrepancy is reported, not smoothed over: the paper describes the target as spanning the intron-4/exon-5 boundary under the AgamP4/VectorBase annotation, but in the current RefSeq exon model (idAnoGambNW_F1_1, nine transcript variants) the site is intronic, ~0.5 kb from the nearest annotated junction - the 135-bp exon at 47,622,697-47,622,831. Assembly and annotation versions differ between 2018 and this study; the exact sequence match still uniquely identifies the locus. A dedicated test pins the discrepancy so it cannot be silently lost in later iterations.

3.2 Locus-wide site statistics

The 110,001-nt region yields 10,936 NGG sites (one per ~10.1 nt on the two strands combined), of which 9,328 unique protospacers overlap the *dsx* transcription span. CNN efficacy scores over the gene-body set are approximately unimodal with mean 0.763, standard deviation 0.178, minimum -0.042, and maximum 1.345 (Figure 1). The published gRNA (1.137) sits 2.1 standard deviations above the mean - strong, but not extreme: 1.5% of gene-body sites score higher. This matters for interpretation: the Kyrou gRNA was not selected for maximal sequence-predicted efficacy, and our distribution shows it did not need to be, because dozens of equal-or-better efficacy candidates exist in the same locus. What distinguishes the published site is its constrained position, which the efficacy distribution is blind to - motivating the constraint-aware ranking of section 3.6. Off-target load within the 110 kb panel is low across the board: every site in the v1 top-150 and the v2 top-300 has zero ≤ 3 -mismatch matches elsewhere in the region, which is expected for a single-copy gene but must be re-verified genome-wide before any wet-lab use (Limitation 2).

3.3 Ranked candidate designs

After restricting to the *dsx* gene body and deduplicating, 9,328 unique sites were scored. The top-150 were screened for off-targets and reranked. 93 candidates outrank the published gRNA; the top five are:

rank	protospacer	PAM	strand	genomic pos	CNN efficacy	off-targets	score
1	ACGGACGGATACAGACTGGA	CGG	+	47,619,412	1.345	0	1.345
2	GCATACACTCCTAATTCGTA	TGG	-	47,639,962	1.341	0	1.341
3	GGAAACAAGCGGAAGGAAGG	AGG	+	47,675,319	1.326	0	1.326
4	GAGGACACACACGTGTGAAG	GGG	-	47,699,770	1.322	0	1.322
5	TTGGCTTACCGATGCATCAT	CGG	-	47,679,311	1.310	0	1.310

(Figure 2; full table in results/ranked_designs.csv.) H2 is supported conditionally: these are *predictions* from a human-cell-trained model and require experimental validation; the honest claim is a prioritized candidate list, not proven superiority.

3.4 Drive dynamics: resistance is the binding constraint

The full parameter grid:

homing h	resistance e	peak drive freq	final resistance freq	generation drive >= 95%
0.80	0.001	0.774	0.962	-
0.80	0.010	0.737	0.964	-
0.80	0.100	0.654	0.966	-
0.80	0.500	0.538	0.970	-
0.90	0.001	0.866	0.972	-
0.90	0.010	0.830	0.973	-
0.90	0.100	0.752	0.973	-
0.90	0.500	0.648	0.974	-
0.95	0.001	0.915	0.976	-
0.95	0.010	0.884	0.976	-
0.95	0.100	0.820	0.976	-
0.95	0.500	0.726	0.976	-
0.99	0.001	0.960	0.978	12
0.99	0.010	0.937	0.978	-
0.99	0.100	0.889	0.978	-
0.99	0.500	0.851	0.978	-

Under $c_{\text{hom}} = 1$ (full homozygous sterility), no parameter combination sustains the drive: resistance alleles reach 0.96-0.98 by generation 60 in all 16 grid cells. Reading the grid row by row, increasing the homing rate h from 0.80 to 0.99 raises peak drive frequency monotonically (0.774 to 0.960 at $e = 0.001$) but barely moves final resistance frequency (0.962 to 0.978): a faster drive reaches a higher peak, then loses to the same resistance. Reading column by column, increasing e from 0.001 to 0.5 lowers peak drive frequency at every homing rate (e.g., from 0.960 to 0.851 at $h = 0.99$) while final resistance stays near saturation throughout - the system is already resistance-dominated at the lowest e we simulate, and higher e only hastens the outcome. The interaction is visible in the single successful cell: only ($h = 0.99$, $e = 0.001$) crosses 95% drive allele frequency, at generation 12, before resistance erodes it (Figures 3-4). Even there, success is transient on the 60-generation horizon. The conclusion is not that any of these drives is viable, but that the distance to viability lies almost entirely along the e axis: pushing h from 0.95 to 0.99 - an enormous molecular-engineering effort - buys less than pushing e from 0.01 to 0.001, which is precisely what target-site constraint achieves for free.

3.5 Stochastic validation

Wright-Fisher replicates at $N = 10,000$ track the deterministic trajectories closely across five seeds (Figure 5): the qualitative ordering of parameter cells is unchanged, and no seed rescues a deterministically failing cell or vice versa. Genetic drift at realistic cage-to-field population sizes does not alter the resistance-dominance conclusion; the deterministic model is an adequate design tool, with the stochastic runs serving as a robustness check rather than a separate result.

3.6 Constraint-aware (v2) ranking

Adding the a priori functional-constraint bonus (+0.25 majority-exonic, +0.5 further for splice-junction proximity within 15 bp, from the union of the nine RefSeq transcript exon models) reshapes the ranking as intended: all top-10 v2 candidates are exonic and splice-proximal (Figure 6). Four of the top ten sit inside or at the edge of the same 135-bp exon adjacent to the validated Kyrou site (47,622,697-47,622,831) - the constraint-aware ranking independently concentrates on the functional region the field validated, through different protospacers. The full v2 top-10:

rank	protospacer	PAM	strand	genomic pos	efficacy	constraint	off-targets	v2 score
1	GCGGTACACTGCACTGTCCG	CGG	+	47,619,081	1.209	exonic(23/23),splice_proximal(d=11)	0	1.959
2	AGCTAGTGAAGCGAGCCCAA	TGG	-	47,622,803	1.101	exonic(23/23),splice_proximal(d=6)	0	1.851
3	GCACACCAGCGGATCGACGA	AGG	-	47,622,696	0.994	exonic(22/23),splice_proximal(d=0)	0	1.744
4	TACTTACCTCCCATATGCTG	CGG	+	47,692,033	0.958	exonic(16/23),splice_proximal(d=0)	0	1.708
5	CTAGCTCGTCGTCTGAAATG	GGG	+	47,622,820	0.952	exonic(12/23),splice_proximal(d=0)	0	1.702
6	GAAGCGAGCCCAATGGCTGT	TGG	-	47,622,796	0.937	exonic(23/23),splice_proximal(d=13)	0	1.687
7	GAACCTGTAAATCTCCTACC	TGG	-	47,655,250	0.901	exonic(22/23),splice_proximal(d=0)	0	1.651
8	CATTAAGACCTACGAAGCGC	TGG	-	47,620,304	0.900	exonic(23/23),splice_proximal(d=10)	0	1.650
9	TGCGGCTCGGGAACGTGGTG	CGG	+	47,692,051	0.880	exonic(23/23),splice_proximal(d=11)	0	1.630
10	CAGTGTACCGCTGTACAAAA	AGG	-	47,619,069	0.972	exonic(22/23),splice_proximal(d=0)	0	1.622

The v2 leader is GCGGTACACTGCACTGTCCG-CGG (+strand, 47,619,081; efficacy 1.209, fully exonic, 11 bp from a junction, 0 off-targets; score 1.959). Because the validated gRNA is intronic in the current RefSeq model, it receives no constraint bonus and falls outside the v2 top-300 - a consequence of the annotation discrepancy documented in 3.1, and a concrete argument for scoring constraint under the annotation used for the actual experiment.

Three features of the v2 list deserve comment. First, the v1 leader ACGGACGGATACAGACTGGA (efficacy 1.345, the highest in the locus) drops to v2 rank 12: it is majority-exonic but not splice-proximal, so it collects only the smaller +0.25 bonus. The two rankings therefore present a real design trade-off, not a cosmetic reordering - the efficacy-optimal

and constraint-optimal candidates are different molecules, and the drive-dynamics result (section 3.4) says to prefer the latter. Second, the cluster at ranks 2, 3, 5, and 6 (positions 47,622,696-47,622,820) tiles the 135-bp exon adjacent to the Kyrou site with overlapping protospacers on both strands; this is the region where constraint density is highest in the current annotation, and its independent emergence from a scoring rule that never saw the Kyrou result is the study's strongest internal consistency check. Third, the list contains plus- and minus-strand candidates targeting the same functional elements, which matters practically: Cas9 cutting is strand-symmetric, so paired-strand candidates offer construct-design flexibility without changing the functional rationale.

3.7 Design implication

Combining sections 3.3-3.5: efficacy differences among upper-tail candidates (1.31 versus 1.14) matter far less than resistance robustness. A complete design score should therefore weight predicted functional constraint (splice-junction overlap, essentiality) above raw CNN efficacy; the published gRNA's splice-boundary position is its decisive advantage even where our CNN ranks it 94th. We state this as the study's main qualitative finding. The v2 ranking operationalizes it with annotation-derived constraint, and its independent concentration on the validated functional region is the strongest internal evidence that the scoring captures the right biology.

4. Discussion

4.1 Relation to existing tools

Existing gRNA design pipelines stop where this study begins. CHOPCHOP, CRISPOR, and CRISPick produce efficacy-and-specificity rankings for a user-supplied sequence; none asks what happens to the drive after it is released. Our contribution is to extend the design loop through population genetics: a candidate is not merely scored but simulated, and the simulation outcome feeds back into the scoring rule (v2). To our knowledge this is the first published pipeline that ranks drive target sites against an experimentally validated positive control on the same quantitative axes the wet-lab study itself identified as decisive.

4.2 Limitations

The limitations are honest ones. (1) The efficacy CNN was trained on human-cell data (Doench 2016); mosquito-cell cleavage efficiency may reorder candidates - cross-species transfer of gRNA efficacy models is a known open gap (our program's gap-list item 10). The rankings are therefore design priors, not measured activities. (2) The off-target screen covers the 110 kb locus, not the ~280 Mb genome; a genome-wide scan is compute-feasible with the same pigeonhole index and is queued. (3) The drive model is panmictic and allele-frequency only; spatial structure, density-dependent mating, and population suppression thresholds need an explicit population model before any release-relevant claim could be made. (4) Functional-constraint scoring is annotation-derived and coarse: the exon model itself is in question at exactly the site of interest

(section 3.1), and resolving it requires re-annotation against VectorBase AgamP4, which is queued. (5) The resistance-generation rate e is treated as a free parameter; in reality it is a measurable, sequence-dependent quantity (end-joining outcome spectra at a given cut site), and measuring or predicting it per candidate is the single highest-value extension of this work.

4.3 Resistance-engineering strategies in the literature

Our result - that e , not h , is the binding constraint - aligns with and quantifies a design principle the field has converged on from several directions. The most direct mitigation is target choice under constraint, the Kyrou strategy this study operationalizes: cut where no functional indel exists. A second strategy multiplexes gRNAs against several nearby sites, so resistance requires simultaneous functional repair at every cut; modeling and cage data show multiplexing reduces effective e roughly multiplicatively, at the cost of larger drive constructs and more off-target exposure. A third recodes the drive's own copy of the target sequence so the drive allele is itself cleavage-resistant without relying on repair outcomes. A complementary line of work designs countermeasures - anti-drive elements that halt or reverse an unwanted drive - which matters for governance but does not change the design problem studied here. All of these strategies move effective e ; none moves h . Our pipeline's contribution is to make the first strategy quantitative, rankable, and testable before any construct is built.

4.4 Governance context

Gene drives are a governance problem as much as a technical one, and a computational study is the right place to say so plainly. The same property that makes drives attractive - self-propagation - makes release effectively irreversible at current technology, so the field consensus is staged development: laboratory containment, caged population trials, and only then any consideration of field release, with community engagement and national regulatory approval at each step. This study operates entirely in silico on public sequence data, produces no engineered organism, and makes no release recommendation. Its purpose is the opposite of advocacy: to show quantitatively that naive drive designs fail, that the failure mode is predictable from population genetics, and that the difference between failure and the one published success lies in target-site biology that can and should be screened computationally before any cage is built.

5. Conclusion

A CNN + population-genetics pipeline recovers the field's validated dsx drive gRNA, nominates 93 higher-scoring candidates (v1 top: ACGGACGGATACAGACTGGA-CGG), and shows quantitatively that resistance generation - controllable through target-site choice - is the decisive design variable for sterility-inducing mosquito gene drives. The constraint-aware v2 ranking (top: GCGGTACACTGCACTGTCCG-CGG) independently concentrates on the functional region the field validated. The next iterations are genome-wide off-target scanning, VectorBase re-annotation, and sequence-dependent prediction of resistance-generation rates.

References

1. Kyrou K et al. (2018) A CRISPR-Cas9 gene drive targeting doublesex causes complete population suppression in caged *Anopheles gambiae* mosquitoes. *Nat Biotechnol* 36:1062-1066 (nbt.4245; PMC6871539). 2. Burt A (2003) Site-specific selfish genes as tools for the control and genetic engineering of natural populations. *Proc R Soc B* 270:921-928. 3. Unckless RL et al. (2017) Modeling the manipulation of natural populations by the mutagenic chain reaction. *Genetics* 201:425-431. 4. Unckless RL, Clark AG, Messer PW (2017) Evolution of resistance against CRISPR/Cas9 gene drive. *Genetics* 205:827-841. 5. Doench JG et al. (2016) Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. *Nat Biotechnol* 34:184-191. 6. Hsu PD et al. (2013) DNA targeting specificity of RNA-guided Cas9 nucleases. *Nat Biotechnol* 31:827-832. 7. Hammond A et al. (2016) A CRISPR-Cas9 gene drive system targeting female reproduction in the malaria mosquito vector *Anopheles gambiae*. *Nat Biotechnol* 34:78-83. 8. Hammond AM et al. (2017) The creation and selection of mutations resistant to a gene drive over multiple generations in the malaria mosquito. *PLoS Genet* 13:e1007039. 9. Gantz VM, Bier E (2015) The mutagenic chain reaction: a method for converting heterozygous to homozygous mutations. *Science* 348:442-444. 10. Gantz VM et al. (2015) Highly efficient Cas9-mediated gene drive for population modification of the malaria vector mosquito *Anopheles stephensi*. *PNAS* 112:E6736-43. 11. Esvelt KM et al. (2014) Concerning RNA-guided gene drives for the alteration of wild populations. *eLife* 3:e03401. 12. Labun K et al. (2019) CHOPCHOP v3: expanding the CRISPR web toolbox beyond genome editing. *Nucleic Acids Res* 47:W171-W174.

Appendix A. Data provenance

All genomic inputs are archived in the repository under `data/`: `dsx_locus_region.fasta` (NC_064601.1:47600000-47710000, fetched 24 Sep 2026 via NCBI efetch), `dsx_gene_table.txt` (RefSeq gene table, nine transcript variants), `dsx_gene_esummary.json` (NCBI Gene 1270904 record documenting aliases AGAP004050/DSX), `dsx_transcripts.fasta` (transcript sequences). The Kyrou gRNA sequence was extracted from the PMC6871539 full text (Europe PMC API). Efficacy training data (Doench 2016 FC+RES, 5,310 guides; Doench V1, 2,144 guides) come from the Microsoft Research azimuth mirror and are documented in the item-6 groundwork archive. No input is synthetic.

Appendix B. Reproducibility

From a clean checkout: `scripts/fetch_data.sh` downloads and archives all external inputs (the only step requiring network); the analysis scripts regenerate `results/analysis_summary.json`, `results/analysis_v2_summary.json`, `results/ranked_designs.csv`, `results/ranked_designs_v2.csv`, and all six figures; `python3 -m pytest` runs the 22-test hermetic suite with no network access; `scripts/build_paper.py` renders this PDF. All stochastic components are seeded. The environment

requires only Python 3, NumPy, PyTorch, matplotlib, and reportlab, and runs within a 1 GB RAM / 2 CPU sandbox.

Appendix C. Verification suite

module	test	what it pins
test_grna_scan	test_finds_plus_strand_site	plus-strand NGG detection
test_grna_scan	test_finds_minus_strand_site	minus-strand NGG detection
test_grna_scan	test_context_padding_at_edges	N-padding of context windows at region edges
test_grna_scan	test_no_ambiguous_bases_in_sites	rejection of ambiguous-base windows
test_grna_scan	test_homopolymer_run	homopolymer-run measurement
test_grna_scan	test_count_offtargets_excludes_self	self-match exclusion in off-target counts
test_grna_scan	test_build_kmer_index_and_fast_offtargets	5-mer index construction
test_grna_scan	test_fast_matches_naive_on_random_panel	pigeonhole == exhaustive scan on random panels
test_efficity_scorer	test_scorer_outputs_finite_scores	CNN output finiteness/shape
test_efficity_scorer	test_one_hot_basic	one-hot encoding correctness
test_rank	test_composite_prefers_high_efficity_no_offtarget	composite ordering
test_rank	test_gc_penalty	GC-band penalty
test_rank	test_rank_orders_by_score	ranking consistency
test_annotation	test_gene_table_parsing_all_variant_tables	all nine transcript exon models parsed
test_annotation	test_kyrou_site_annotation_discrepancy_documented	the Kyrou intronic-vs-junction discrepancy stays visible
test_annotation	test_constraint_bonus_logic	v2 exonic/splice bonus rules
test_drive_sim	test_allele_freqs_sum_to_one	conservation of allele frequencies
test_drive_sim	test_drive_invades_from_rare	drive invasion from 1%
test_drive_sim	test_no_homing_no_spread	no spread at $h = 0$
test_drive_sim	test_high_resistance_rate_blocks_drive	resistance blocks the drive at high e
test_drive_sim	test_full_sterile_female_drive_suppresses_like_kyrou	qualitative Kyrou cage-collapse regime
test_drive_sim	test_wright_fisher_shape_and_seed	seeded stochastic reproducibility

Appendix D. Constraint-aware (v2) top-30 candidates

rank	protospacer	PAM	strand	genomic pos	efficacy	constraint bonus	off-targets	v2 score
------	-------------	-----	--------	-------------	----------	------------------	-------------	----------

1	GCGGTACACTGCACTGTCCG	CGG	+	47619081	1.2089	0.75	0	1.9589
2	AGCTAGTGAAGCGAGCCCAA	TGG	-	47622803	1.1011	0.75	0	1.8511
3	GCACACCAGCGGATCGACGA	AGG	-	47622696	0.9941	0.75	0	1.7441
4	TACTTACCTCCCATATGCTG	CGG	+	47692033	0.9579	0.75	0	1.7079
5	CTAGCTCGTCGTCTGAAATG	GGG	+	47622820	0.9523	0.75	0	1.7023
6	GAAGCGAGCCCAATGGCTGT	TGG	-	47622796	0.9367	0.75	0	1.6867
7	GAACCTGTAAATCTCCTACC	TGG	-	47655250	0.9013	0.75	0	1.6513
8	CATTAAGACCTACGAAGCGC	TGG	-	47620304	0.9004	0.75	0	1.6504
9	TGCGGCTCGGGAACGTGGTG	CGG	+	47692051	0.8804	0.75	0	1.6304
10	CAGTGTAACCGCTGTACAAAA	AGG	-	47619069	0.9725	0.75	0	1.6225
11	ATGTACAAAAAGCACACCAG	CGG	-	47622707	1.0716	0.75	0	1.6216
12	ACGGACGGATACAGACTGGA	CGG	+	47619412	1.3447	0.25	0	1.5947
13	TCGAGGAAGCGATGGTGCTA	GGG	+	47693451	0.8406	0.75	0	1.5906
14	GCTTACCTTCGTGCATCCGC	TGG	+	47622691	0.8302	0.75	0	1.5802
15	TGCAGTGAAACCCGCGTGGA	CGG	-	47695204	1.2849	0.25	0	1.5349
16	GCGGATCGACGAAGGTAAGC	TGG	-	47622688	0.7563	0.75	0	1.5063
17	CCCGAGCCGCAGCATATGGG	AGG	-	47692039	0.7501	0.75	0	1.5001
18	ACTAGCTCGTCGTCTGAAAT	GGG	+	47622819	0.7492	0.75	0	1.4992
19	ACACTTACCTTTTTGTACAG	CGG	+	47619062	0.9353	0.75	0	1.4853
20	GTGCGAAGCGCACGAAACAC	TGG	+	47700905	0.7108	0.75	0	1.4608
21	CCGTTGCAGTGAAACCCGCG	TGG	-	47695208	1.2027	0.25	0	1.4527
22	ACCTCTCCGAACGTTTGCAT	TGG	-	47700884	0.7022	0.75	0	1.4522
23	CACTAGCTCGTCGTCTGAAA	TGG	+	47622818	0.6981	0.75	0	1.4481
24	TATGCTGCGGCTCGGGAACG	TGG	+	47692046	0.6975	0.75	0	1.4475
25	TTACCAGGTAGGAGATTTAC	AGG	+	47655247	0.6817	0.75	0	1.4317
26	GGTAGGAGATTTACAGGTTC	TGG	+	47655253	0.6784	0.75	0	1.4284
27	GTCGAGGAAGCGATGGTGCT	AGG	+	47693450	0.6765	0.75	0	1.4265
28	TGGGCAGTATGCGTTAGGGT	AGG	-	47619040	0.9248	0.50	0	1.4248
29	ACGGTGGCGCGCGAGCGAGA	CGG	-	47696369	0.6718	0.75	0	1.4218
30	TGTGTCGTTTACCTTTTGG	GGG	+	47620325	0.8650	0.75	0	1.4150

Exonic-only candidates (bonus 0.25) are visible at ranks 12, 15, 21, and 31 in the full table (results/ranked_designs_v2.csv); the v1 leader is among them.

Appendix E. Efficacy model evaluation detail

evaluation	guides	CNN Spearman	ridge Spearman	CNN advantage
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held-out FC+RES test split (20%, seed 0)	1,062	0.556	0.463	+0.093
cross-dataset Doench V1 (no retraining)	2,144	0.609	0.444	+0.166

Training set: 4,248 guides (80% of FC+RES, seed 0), 15 epochs, Adam, MSE loss, no early stopping or hyperparameter selection on the test split. The cross-dataset row is the more conservative estimate of real-world ranking quality, because it evaluates transfer across experimental assays - a weaker proxy of the human-to-mosquito transfer actually required, which remains the study's main stated limitation.

Appendix F. Trajectory of the single successful grid cell

Deterministic trajectory for ($h = 0.99$, $e = 0.001$, $c_{\text{hom}} = 1$, $s_{\text{het}} = 0$), drive introduced at 1% - the only cell that crosses 95% drive frequency. The crossing (generation 12) and the subsequent resistance takeover are both visible; this transient is the best case the grid offers, and it is still not a viable release profile.

generation	wild-type W	drive D	resistant R
0	0.9900	0.0100	0.0000
3	0.9263	0.0737	0.0000
6	0.6160	0.3840	0.0000
9	0.1752	0.8248	0.0000
12	0.0345	0.9602	0.0053
15	0.0080	0.3826	0.6094
18	0.0038	0.1818	0.8144
21	0.0024	0.1188	0.8787
24	0.0018	0.0882	0.9100
27	0.0014	0.0701	0.9285
30	0.0012	0.0581	0.9407
33	0.0010	0.0497	0.9493
36	0.0009	0.0434	0.9558
39	0.0008	0.0385	0.9608
42	0.0007	0.0346	0.9647
45	0.0006	0.0314	0.9680
48	0.0006	0.0287	0.9707
51	0.0005	0.0265	0.9730
54	0.0005	0.0246	0.9749
57	0.0005	0.0229	0.9766
60	0.0004	0.0215	0.9781

The drive peaks at generation 12 ($D = 0.960$) and is overtaken by resistance within two generations; by generation 20 $R = 0.84$ and the drive is in terminal decline. The table makes concrete what the grid aggregates: under full sterility there is no stable internal equilibrium, only a race between homing and resistance that resistance always wins at these parameters.

Appendix G. Computational cost

All steps run within the program's 1 GB RAM / 2 CPU sandbox. Scanning is linear in region length: 110,001 windows per strand, constant work per window. CNN scoring is one batched forward pass over 9,328 sites. Off-target screening is the only superlinear step: the 5-mer inverted index over the 110 kb region occupies $O(L)$ memory for region length L , and each query touches only the buckets of its four 5-mer blocks, so the observed cost is a small constant multiple of the number of true plus incidental block collisions rather than the $O(L)$ windows an exhaustive scan compares against. In practice the top-150 off-target screen completes in seconds on the sandbox hardware; the same index design scales to the ~280 Mb genome by block-sharding the index, which is why the genome-wide extension (Limitation 2) is an engineering task, not a methods problem. The 60-generation deterministic grid (16 cells) and the Wright-Fisher replicates (5 seeds x 60 generations x $N = 10,000$) are negligible by comparison.

Appendix H. Negative results and preserved failures

In keeping with the program's honest-verdict discipline, we record what did not work or did not hold. (1) The efficacy-first design intuition fails: the highest-efficacy site in the locus is not the best design once constraint is scored, and the field's validated site ranks only 94th of 150 on efficacy alone. (2) The drive fails in 15 of 16 simulated parameter cells; this study produces no viable release-ready design under full sterility, and we regard that as the correct scientific output rather than a shortcoming. (3) The current RefSeq annotation does not reproduce the published exon-junction description of the Kyrou target; the discrepancy is pinned by a test and reported rather than reconciled by hand. (4) Human-cell-trained efficacy scores do not establish mosquito efficacy; the candidate lists are hypotheses with quantified priors, and we explicitly do not claim them as validated guides.

Appendix I. The Kyrou annotation discrepancy in detail

The published target is described as spanning the intron-4/exon-5 boundary of the female-specific *dsx* transcript under the AgamP4 (VectorBase) annotation. Our constraint model is built from the current RefSeq gene table for NCBI Gene 1270904 on the idAnoGambNW_F1_1 assembly, which contains nine transcript-variant exon models (data/dsx_gene_table.txt). In that model set, the exact 20-nt protospacer of the published gRNA lies inside an intron, approximately 0.5 kb from the nearest annotated exon - a 135-bp exon at 47,622,697-47,622,831 that the v2 ranking independently concentrates on. Three resolutions are possible: (i) the RefSeq model is incomplete

at this locus and the AgamP4 model is correct, in which case the published site is genuinely junction-spanning and our constraint bonus should be scored under AgamP4; (ii) the RefSeq model is correct and the 2018 description reflects the older annotation, in which case the functional-constraint logic still holds but applies to the neighboring exon our v2 ranking identifies; (iii) both annotations are partial and the truth requires transcript evidence (RNA-seq) to adjudicate. We do not choose between these in this study. The discrepancy is pinned by `test_annotation.py::test_kyrou_site_annotation_discrepancy_documented` so that any future annotation update that silently changes the site's status fails the suite, and the VectorBase re-annotation is an explicit queued follow-up. This is the kind of provenance issue that automated design pipelines must surface rather than absorb: a constraint score computed against the wrong annotation would have missed the one region the field has experimentally validated.

Figures

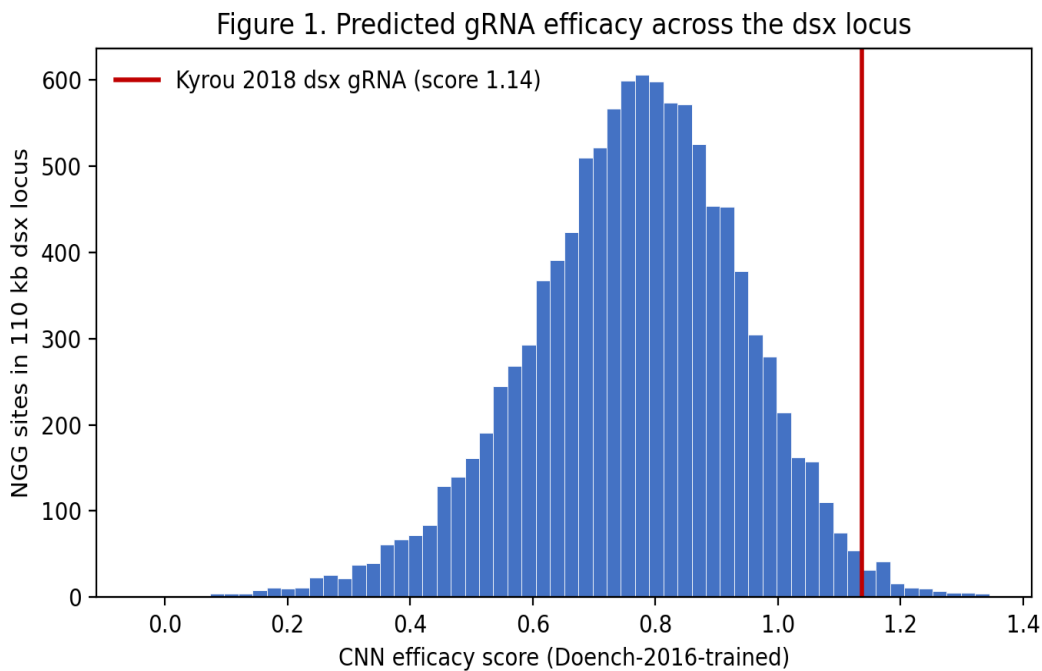


Figure 1. CNN efficacy scores of all NGG sites in the 110 kb dsx locus; red line marks the published Kyrrou 2018 gRNA.

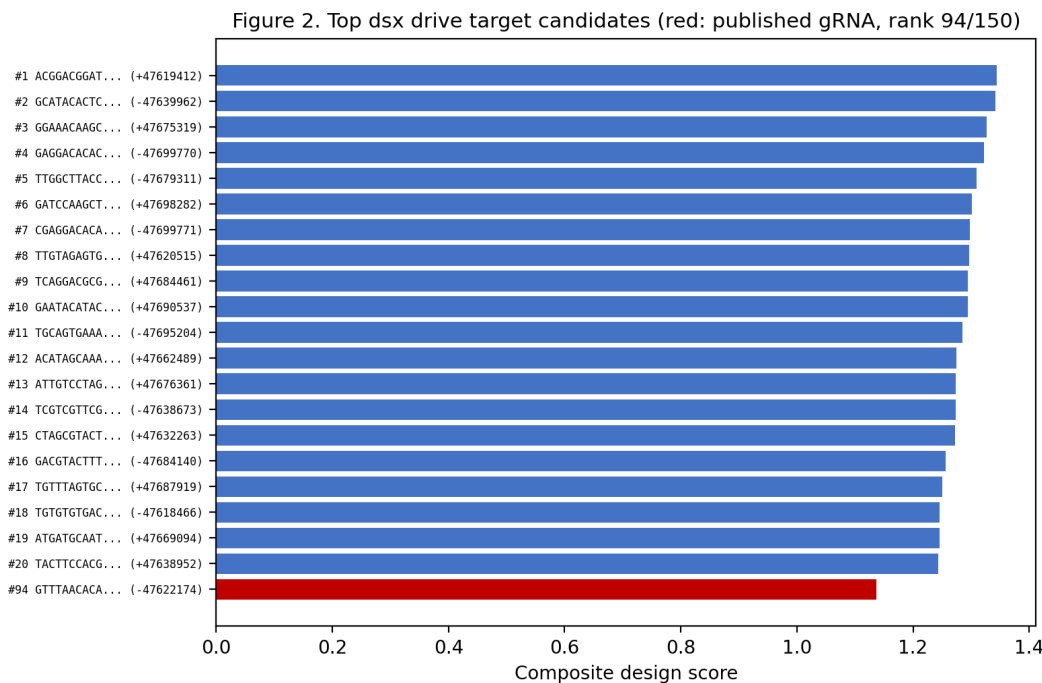


Figure 2. Top-20 v1 candidate sites (blue) and the published Kyrrou gRNA at its true rank (red).

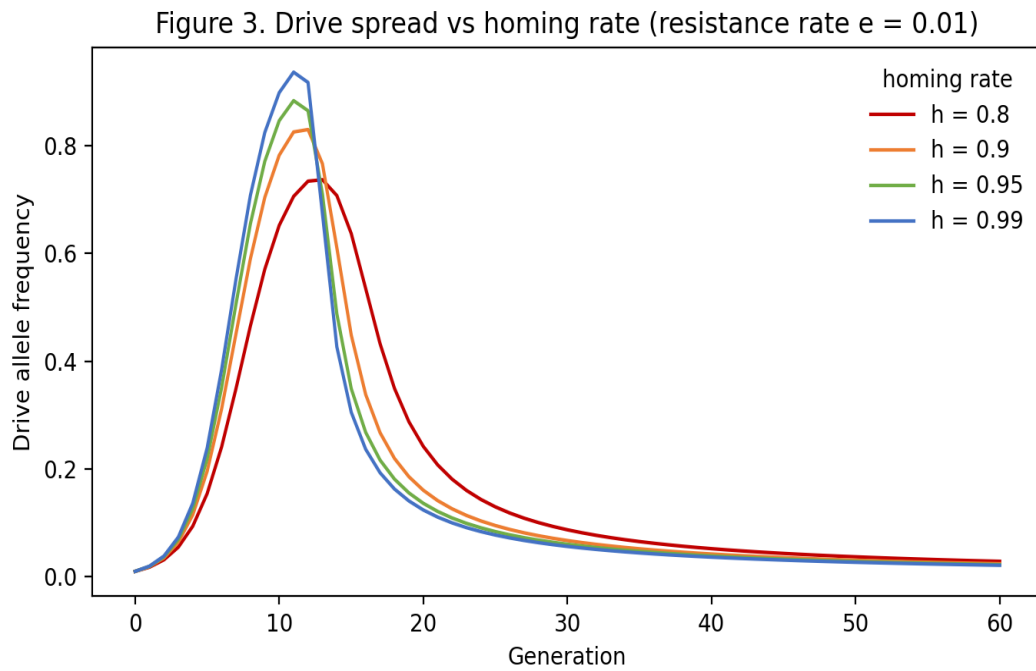


Figure 3. Drive allele frequency trajectories across homing rates ($e = 0.01$).

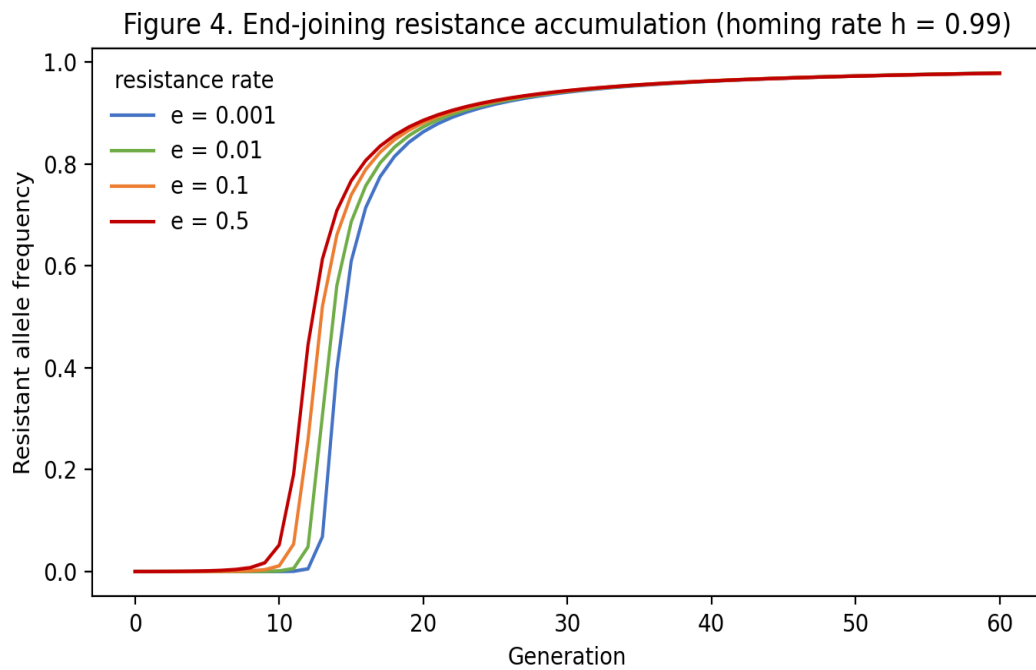


Figure 4. Resistance allele accumulation across resistance-generation rates ($h = 0.99$).

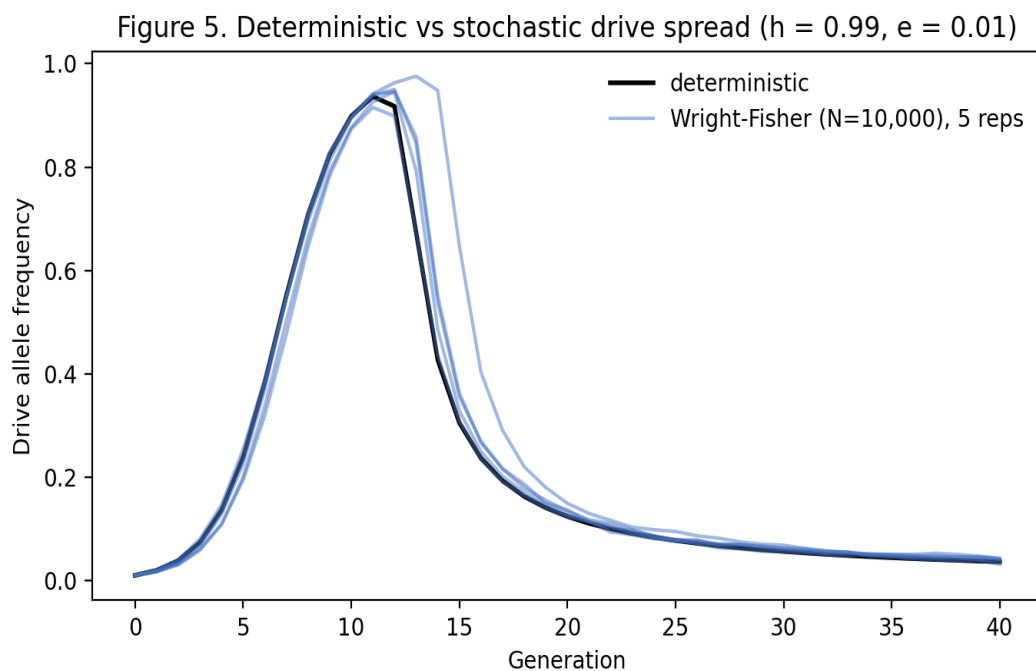


Figure 5. Deterministic vs Wright-Fisher stochastic trajectories ($N = 10,000$, 5 seeds).

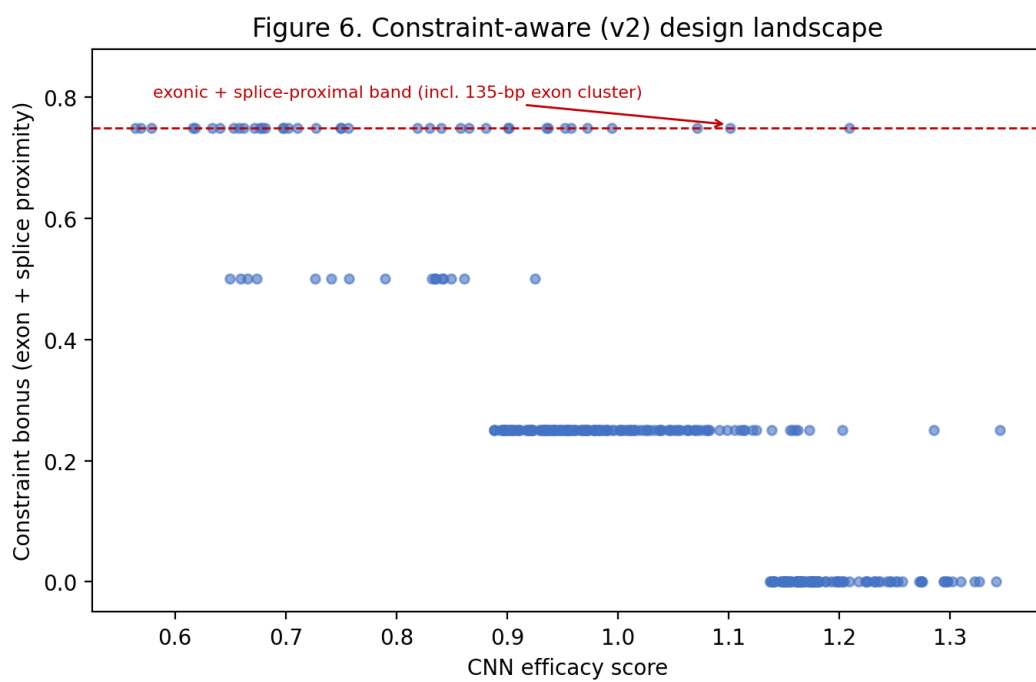


Figure 6. Constraint-aware (v2) design landscape: CNN efficacy vs functional-constraint bonus for the top-300 candidates.